

CNN Occlusion Sensitivity Analysis Report

Abstract

This project investigates how convolutional neural networks make image classification decisions through occlusion sensitivity analysis. Using a pre-trained VGG16 model, portions of two images ("peppers.png" and "llama.jpg") were systematically masked with gray squares of varying sizes. Changes in classification confidence and predicted classes were analyzed to identify the image regions most important to object recognition. The results provide insight into CNN feature extraction and model interpretability.

Introduction

This project investigates the classification behavior of a pre-trained VGG16 Convolutional Neural Network (CNN) using two sample images, *peppers.png* and *llama.jpg*. The methodology is inspired by the work of Zeiler and Fergus in *Visualizing and Understanding Convolutional Networks*, which introduced visualization techniques for understanding how CNNs learn and utilize image features. Their research demonstrated that visualizing network activations and systematically occluding portions of an image can provide valuable insight into the internal decision-making process of deep learning models.

To explore these concepts, occlusion sensitivity analysis was performed by systematically masking regions of each image with gray squares of varying sizes and evaluating the resulting changes in classification predictions. The images were processed through a pre-trained VGG16 network, and the effects of different mask locations and sizes were analyzed using classification confidence scores and prediction outputs.

The objective of this project is to identify which image regions contribute most strongly to object recognition and to better understand how convolutional neural networks interpret visual information. By observing how model predictions change when important features are hidden, this study provides insight into CNN feature extraction, model interpretability, and Explainable Artificial Intelligence (XAI) techniques.

Experimental Setup

The experiment was implemented in Python using TensorFlow, Keras, NumPy, Matplotlib, and the Python Imaging Library (PIL). A pre-trained VGG16 convolutional neural network, trained on the ImageNet dataset, was used to perform image classification. Two test images, *peppers.png* and *llama.jpg*, were selected as inputs for the analysis.

Each image was first converted to RGB format and resized to 224×224 pixels to match the input requirements of the VGG16 architecture. Occlusion sensitivity analysis was then performed by systematically applying gray square masks across the image. Three mask sizes were evaluated: 32×32 , 64×64 , and 128×128 pixels.

For each mask size, the gray square was moved across the image at fixed coordinate intervals until all possible image regions had been occluded once. After each mask was applied, the modified image was preprocessed and passed through the VGG16 network for inference. The top five predicted classes and their corresponding confidence scores were recorded for every masked image.

To evaluate the impact of occlusion on classification behavior, a Mean Feature Sign Change analysis was performed by tracking changes in the network's top predictions as the mask location varied. The resulting data were visualized using Mean Feature Sign Change versus Iteration graphs for each mask size. Additionally, classification confidence values were collected and displayed in the Accuracy of Mask Size charts to compare the effects of different occlusion levels on network performance.

This experimental procedure provided insight into which image regions contributed most strongly to object recognition and how classification confidence changed when important visual features were hidden.

Results

The occlusion sensitivity analysis was performed on both *peppers.png* and *llama.jpg* using gray square masks of size 32×32 , 64×64 , and 128×128 pixels. For each mask size, the mask was systematically moved across the image, and the resulting classification predictions were recorded using the pre-trained VGG16 network.

Figures shown in Tables 1 and 3 illustrate masked images generated during the experiment, along with the corresponding Mean Feature Sign Change versus Iteration plots. The Mean Feature Sign Change plots demonstrate how the network's predictions varied as different image regions were occluded. Smaller mask sizes yielded more iterations because more image locations were evaluated, resulting in a finer-grained analysis of feature sensitivity. Larger mask sizes required fewer iterations and produced broader changes in classification behavior.

Tables 2 and 4 present the classification confidence results for each mask size. The confidence plots indicate that *llama.jpg* generally maintained higher classification confidence than *peppers.png* throughout the experiment. This suggests that the features used to recognize the llama remained more distinguishable under occlusion, whereas the peppers image exhibited greater sensitivity to masked regions.

Overall, the results show that occluding different portions of an image can significantly affect classification confidence and predicted classes. The generated visualizations provide insight into which image regions contribute most strongly to the CNN's decision-making and demonstrate the usefulness of occlusion sensitivity analysis for understanding deep learning models.

Graphs

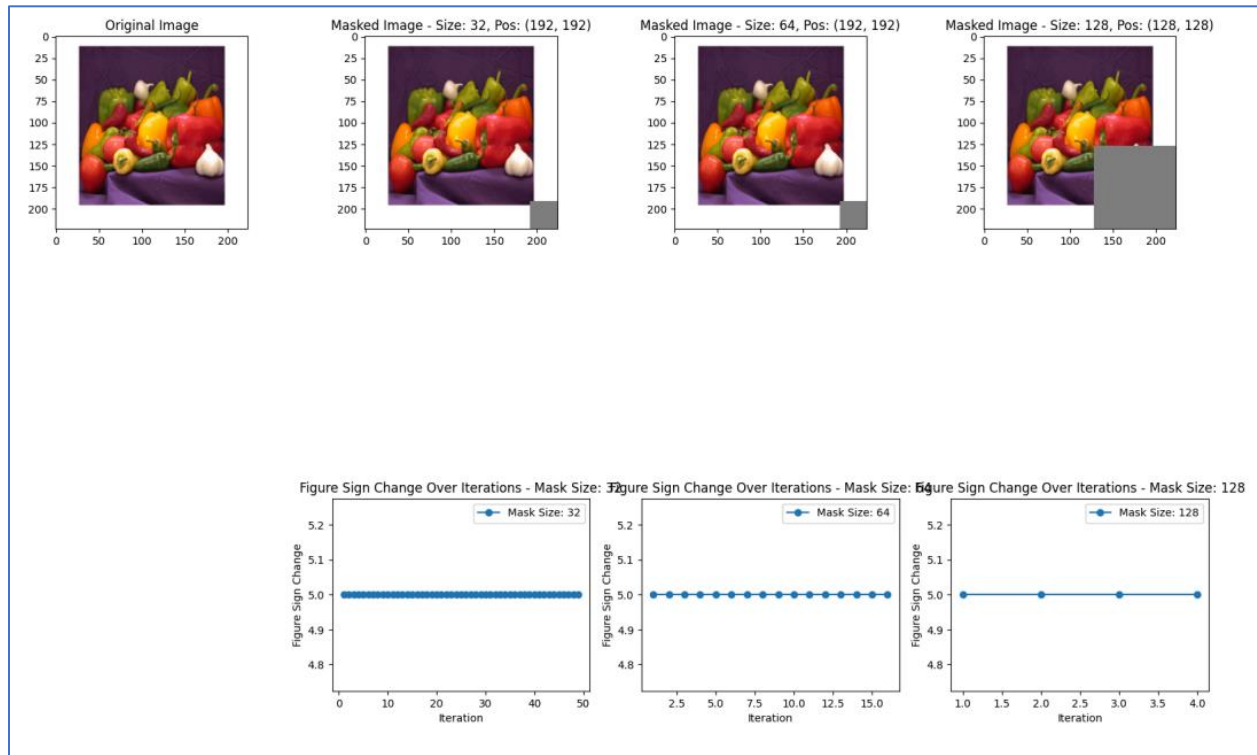


Table 1. (Top Left) Original “peppers.png” Image.
 (Top Three to the Right) Sample of Masked Images.
 (Bottom Three) “peppers.png” Mean Feature Sign Change versus Iteration graphs.

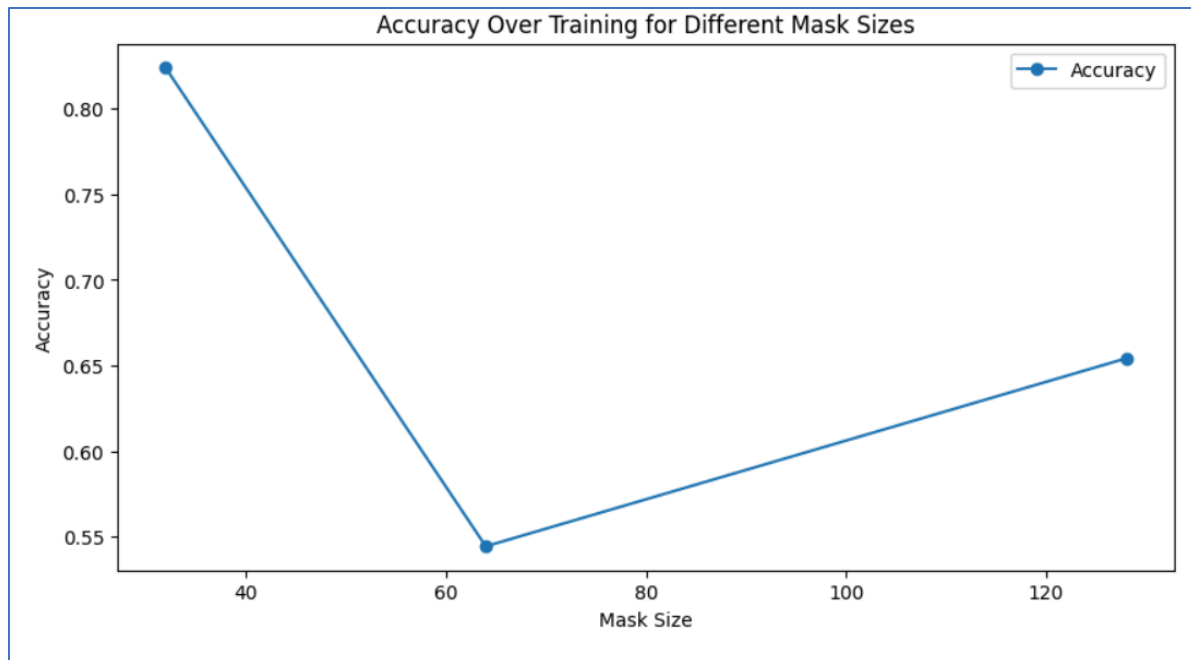


Table 2. Accuracy of Mask Size Chart of the “peppers.png” Image.

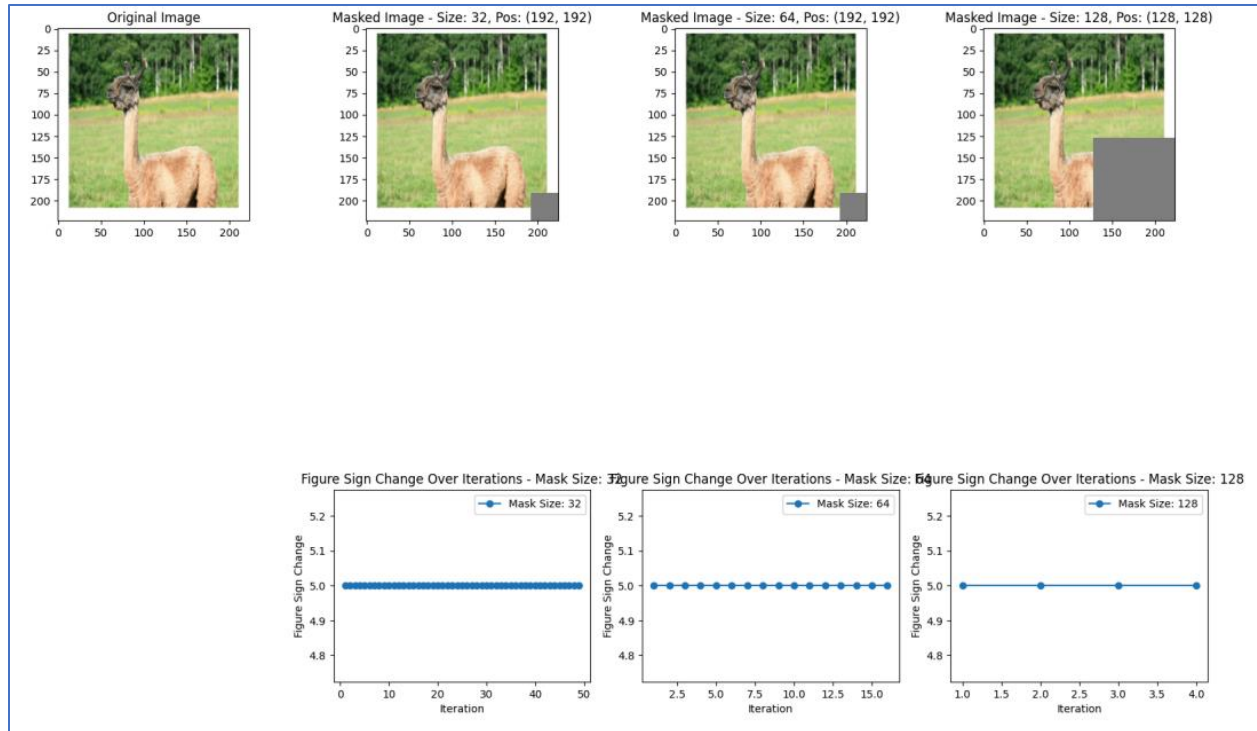


Table 3. (Top Left) Original "llama.jpg" Image.
 (Top Three to the Right) Sample of Masked Images.
 (Bottom Three) "llama.jpg" Mean Feature Sign Change versus Iteration graphs.

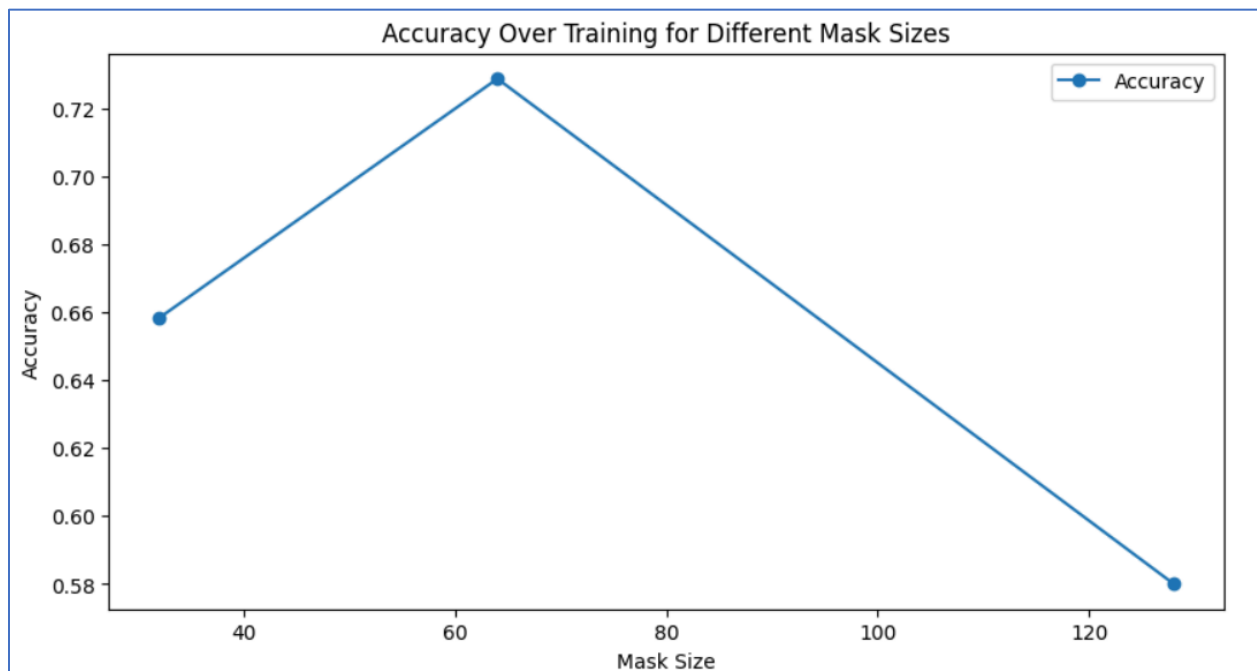


Table 4. Accuracy of Mask Size Chart of the "llama.jpg" Image.

Observations

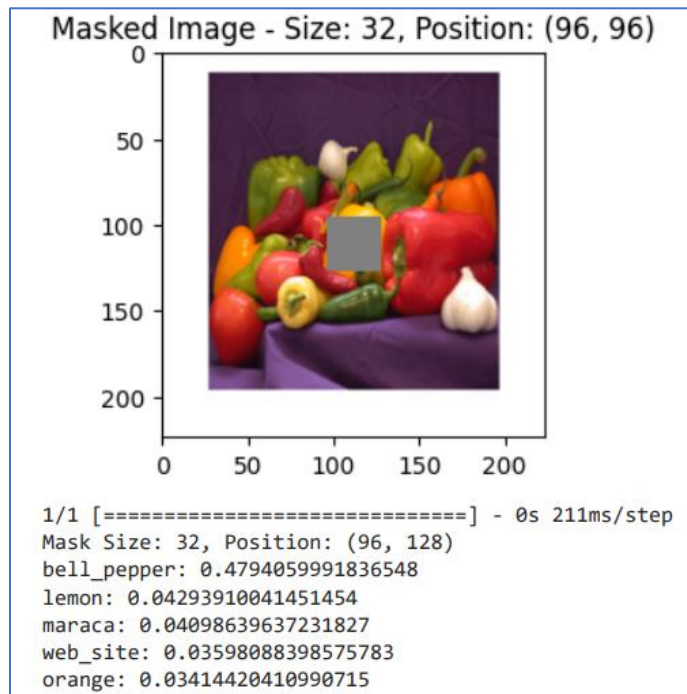


Image 5. Masked Image of “peppers.png” being processed.

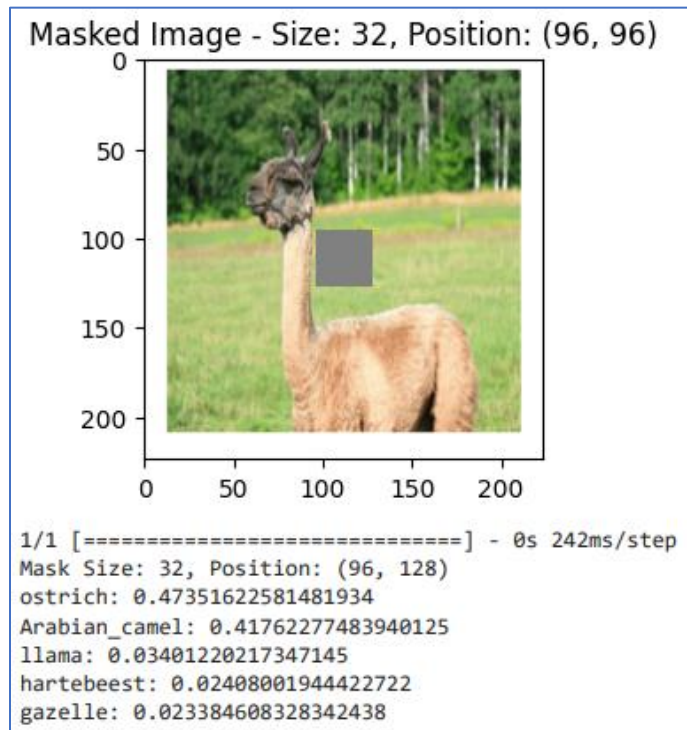


Image 6. Masked Image of “llama.jpg” being processed.

The occlusion sensitivity analysis revealed several important characteristics of CNN-based image classification. First, the two test images exhibited different classification behavior when portions of each image were masked. The *peppers.png* image contained multiple objects, colors, and textures, resulting in greater variability in classification confidence as different regions were occluded. In contrast, the *llama.jpg* image contained a more distinct primary object, allowing the network to maintain relatively stable classification confidence even when portions of the image were masked.

The masked image visualizations in Images 5 and 6 showed how classification confidence changed as different regions of the image were masked. Occluding certain areas produced larger changes in network predictions than others, indicating that the CNN relied more heavily on specific visual features for object recognition. These observations provide insight into the network's internal decision-making process and illustrate how occlusion sensitivity analysis can be used as an Explainable AI (XAI) technique.

The Mean Feature Sign Change analysis also revealed the effect of mask size on the experiment. Smaller masks (32×32 pixels) required more iterations because more image locations were evaluated. This provided a finer-grained analysis of feature importance and enabled examination of individual image regions in greater detail. Larger masks (64×64 and 128×128 pixels) required fewer iterations but produced broader disruptions to image content, resulting in a more generalized view of classification sensitivity.

Overall, the experiment demonstrated that image occlusion can significantly influence classification confidence and predicted classes. The results highlight the importance of specific image features in CNN-based object recognition and show how visualization techniques can improve understanding of deep learning model behavior.

Conclusion

This experiment demonstrated how occlusion sensitivity analysis can be used to understand CNN decision-making and visualize the image features most important for object recognition. By systematically masking portions of the input images and observing changes in classification confidence, it was possible to identify regions that contributed most strongly to the network's predictions. The mean feature sign change plots showed that smaller mask sizes (32 pixels) resulted in more iterations and provided finer-grained insight into model behavior, whereas larger mask sizes (128 pixels) produced fewer iterations and broader disruptions to classification performance. Additionally, the classification confidence plots indicated that *llama.jpg* exhibited higher confidence peaks than *peppers.png*. Overall, the results demonstrate how visualization techniques and occlusion analysis can improve understanding of deep learning models, support model debugging, and contribute to Explainable AI (XAI) research.

References

- [1] M. D. Zeiler and R. Fergus, "Visualizing and Understanding Convolutional Networks," in Proceedings of the European Conference on Computer Vision (ECCV), Zurich, Switzerland, Sep. 2014, pp. 818–833, doi: 10.1007/978-3-319-10590-1_53.
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